Week 17 Progress Report

Details

- Name of Student: Tirthak Girish Likhar
- PRN: 22070521041
- Name of Industry: SS Infotech Pvt. Ltd.
- **Reporting Week: (27/04/2026 to 01/05/2026)**
- Institute Mentor Name: Dr. Shreyas Hole
- Industry Mentor Name: Mr. Viraj Patle

Overview of the Week

- Week 17 focused on building a custom security monitoring dashboard in the Wazuh Dashboard using OpenSearch Dashboards visualization engine. The goal was to aggregate data from all simulations conducted throughout the internship into a single unified view providing a real-time, consolidated security posture overview of the SOC lab environment.

- Three custom visualizations were created from scratch using the wazuh-alerts index as the data source a vertical bar chart showing alert distribution by rule severity level, a pie chart showing the top triggered rule descriptions, and a data table displaying the top MITRE ATT&CK tactics detected across all agents. The dashboard was saved as 'SOC Lab - Security Overview'.

- The exercise demonstrated how SOC analysts use SIEM dashboards to quickly identify patterns, prioritize high-severity alerts, and map detections to threat intelligence frameworks such as MITRE ATT&CK without having to manually query raw log data.

Objectives of the Week

- To create a custom Wazuh dashboard using OpenSearch Dashboards with at least three meaningful security visualizations.

- To build a vertical bar chart visualizing alert distribution across rule severity levels (rule.level) from the wazuh-alerts index.

- To build a pie chart visualizing the top 5 most frequently triggered rule descriptions (rule.description) from the wazuh-alerts index.

- To build a data table visualizing the top 5 MITRE ATT&CK tactics (rule.mitre.tactic) by alert count from the wazuh-alerts index.

- To save the completed dashboard as 'SOC Lab - Security Overview' for use during the Week 18 final demo recording.

- To analyse the dashboard output and document key observations about the SOC lab's alert distribution and MITRE tactic coverage.

Key Accomplishments

- Successfully created a new custom dashboard 'SOC Lab - Security Overview' using OpenSearch Dashboards within the Wazuh interface.

- Built Visualization 1 - Alerts by Rule Level: Vertical bar chart using wazuh-alerts index with rule.level on X-axis (Terms aggregation, Top 10, Descending). Confirmed alert concentration at rule level 5 with smaller counts at levels 12 and 13.

- Built Visualization 2 - Top Rules by Description: Pie chart using wazuh-alerts index with rule.description split slices (Terms, Top 5). Top rules: Windows System error events, NVIDIA LocalSystem Container terminated unexpectedly, Multiple System error events, Windows logon success, Registry Value Entry deleted.

- Built Visualization 3 - Top MITRE Tactics: Data table using wazuh-alerts index with rule.mitre.tactic split rows (Terms, Top 5). Results: Defense Evasion (1,151), Impact (444), Privilege Escalation (400), Initial Access (359), Persistence (359).

- All three panels added to the dashboard simultaneously - dashboard saved as 'SOC Lab - Security Overview' and ready for the Week 18 final demo.

- Confirmed that Defense Evasion is the dominant MITRE tactic with 1,151 alerts - indicating registry changes, process activity, and system configuration events from the Windows endpoint are the primary driver of alert volume.

Issues Faced & Resolutions – Week 17

- **1. No existing saved visualizations available when clicking Add panel**

  - **Issue:** When attempting to add panels to the new dashboard via the Add button, the Add panels drawer showed 'No matching objects found' — no pre-existing Wazuh visualizations were saved in the OpenSearch Dashboards instance.

  - **Resolution:** Used the Create new button within the Add panels drawer to build each visualization from scratch. Selected the appropriate chart type, chose wazuh-alerts-* as the data source, and configured Buckets/Aggregations manually for each panel.

Tools Used – Week 17

Tool / Technology	Purpose
Wazuh Dashboard v4.7.5	Primary interface for building and viewing the custom dashboard
OpenSearch Dashboards	Underlying visualization engine used by Wazuh for custom panels
wazuh-alerts-* index	Data source for all three visualizations contains all Wazuh alert data
Vertical Bar Chart	Visualizing alert count distribution by rule.level
Pie Chart	Visualizing top triggered rules by rule.description
Data Table	Displaying top MITRE ATT&CK tactics by alert count (rule.mitre.tactic)
Wazuh Manager v4.7.5	SIEM backend generating alerts ingested into the dashboard
VirtualBox 7.x	Hypervisor hosting the Ubuntu Server VM
Ubuntu Server 24.04	Wazuh Server — Manager, Indexer, and Dashboard (192.168.1.6)

Conclusion of Week 17

Week 17 successfully delivered a custom SOC monitoring dashboard that aggregates data from all simulations conducted throughout the 18-week internship into a single, unified view. The three visualizations alert severity distribution, top triggered rules, and top MITRE ATT&CK tactics provide exactly the kind of at-a-glance situational awareness that SOC analysts rely on during shift handovers and incident triage.

The dominance of Defense Evasion as the top MITRE tactic (1,151 alerts) reflects the high volume of registry and system configuration events from the Windows 11 endpoint, which aligns with the FIM and system monitoring modules active in the Wazuh agent.

With the dashboard complete, the SOC lab now has a fully operational, multi-layered security monitoring environment covering threat detection, attack simulation, active response, compliance assessment, and visual reporting ready for the final demo recording in Week 18.

Dashboard Visualizations Summary

Visualization	Type	Data Source / Field
Alerts by Rule Level	Vertical Bar Chart	wazuh-alerts-* | rule.level (Terms, Top 10)
Top Rules by Description	Pie Chart	wazuh-alerts-* | rule.description (Terms, Top 5)
Top MITRE Tactics	Data Table	wazuh-alerts-* | rule.mitre.tactic (Terms, Top 5)

Top MITRE ATT&CK Tactics — Data Table Results

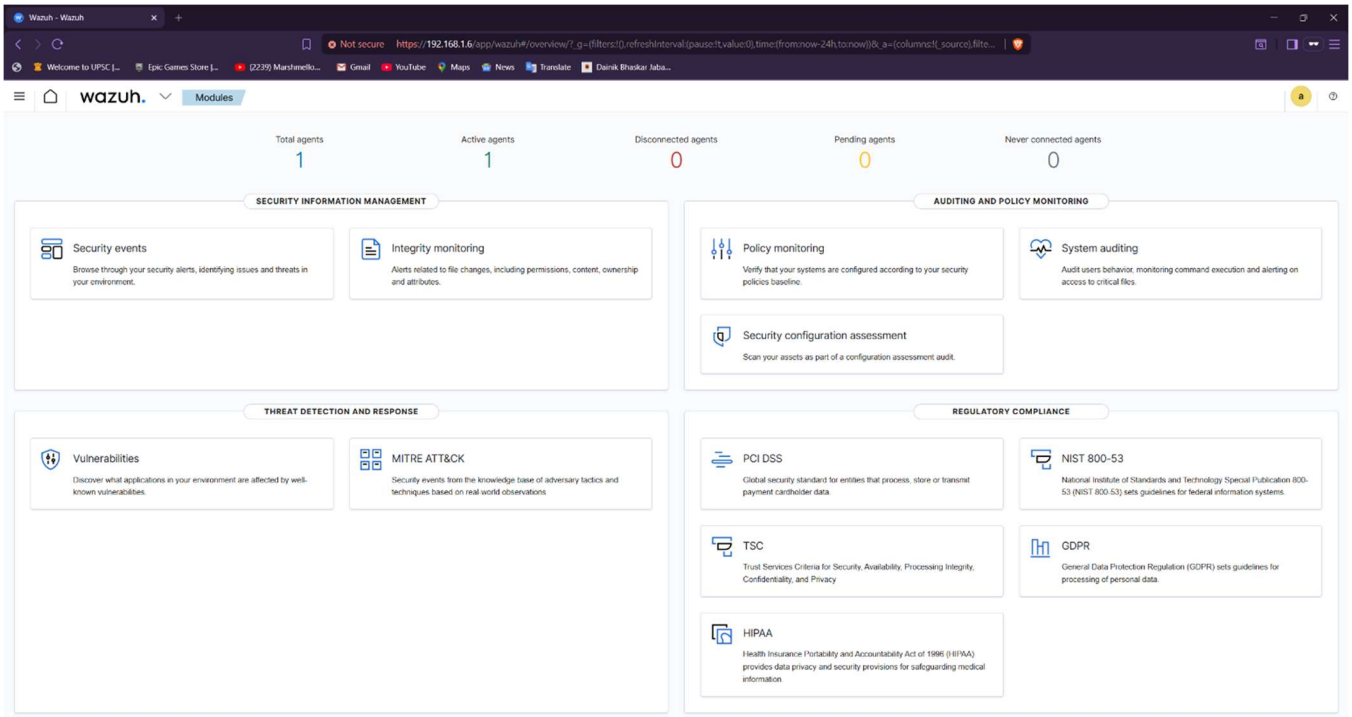
MITRE Tactic	Alert Count	Significance
Defense Evasion	1,151	Highest techniques used to avoid detection (e.g. registry changes, process injection)
Impact	444	Actions that disrupt availability or integrity of systems
Privilege Escalation	400	Attempts to gain higher-level permissions on the endpoint
Initial Access	359	Techniques to gain entry into the network or system
Persistence	359	Techniques used to maintain foothold across reboots

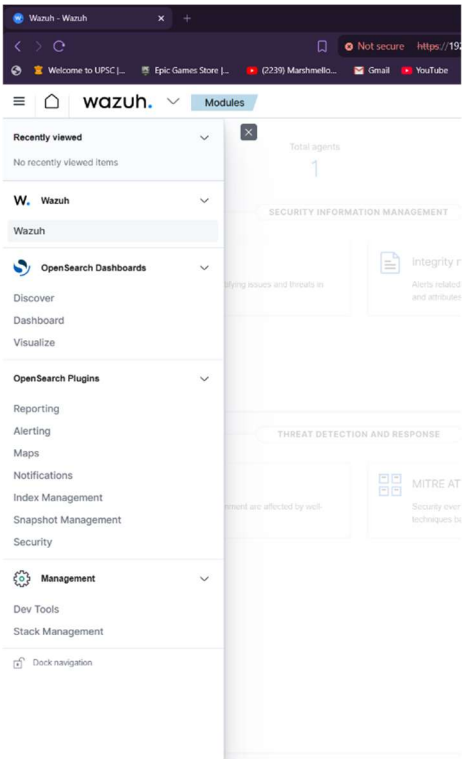
Top Rules by Description — Pie Chart Results

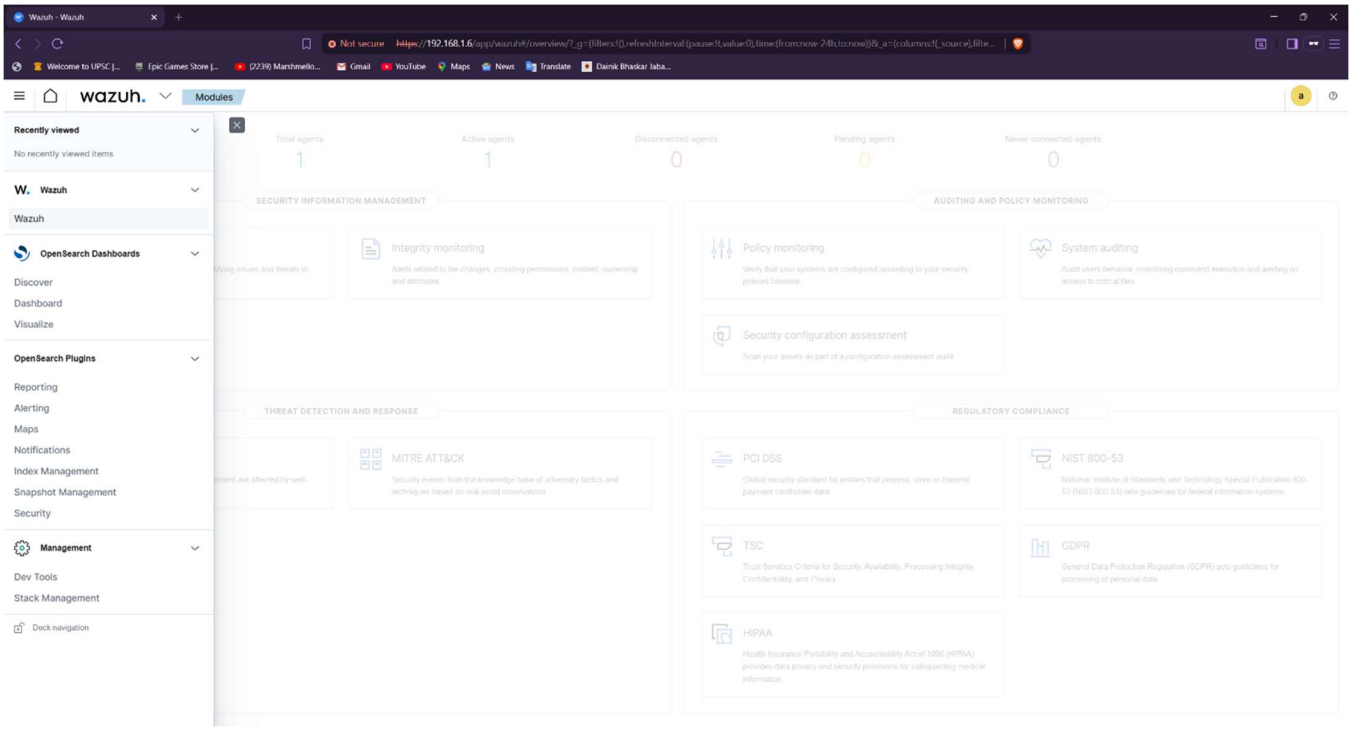
Rule Description	Count (approx.)
Windows System error event	Highest
NVIDIA LocalSystem Container terminated unexpectedly	High
Multiple System error events	Medium
Windows logon success	Low-Medium
Registry Value Entry deleted	Low

Screenshots of Activities (if Any)

[OpenSearch Dashboards visualization editor, Alerts by Rule Level bar chart, Top Rules by Description pie chart, Top MITRE Tactics data table, final 'SOC Lab - Security Overview' dashboard with all 3 panels]



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Modules
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Wazuh
OpenSearch Dashboards
Discover
Dashboard
Visualize
OpenSearch Plugins
Reporting
Alerting
Maps
Notifications
Index Management
Snapshot Management
Security
Management
Dev Tools
Stack Management
Dock navigation
Total agents
1
Active agents
1
Disconnected agents
0
Pending agents
0
Never connected agents
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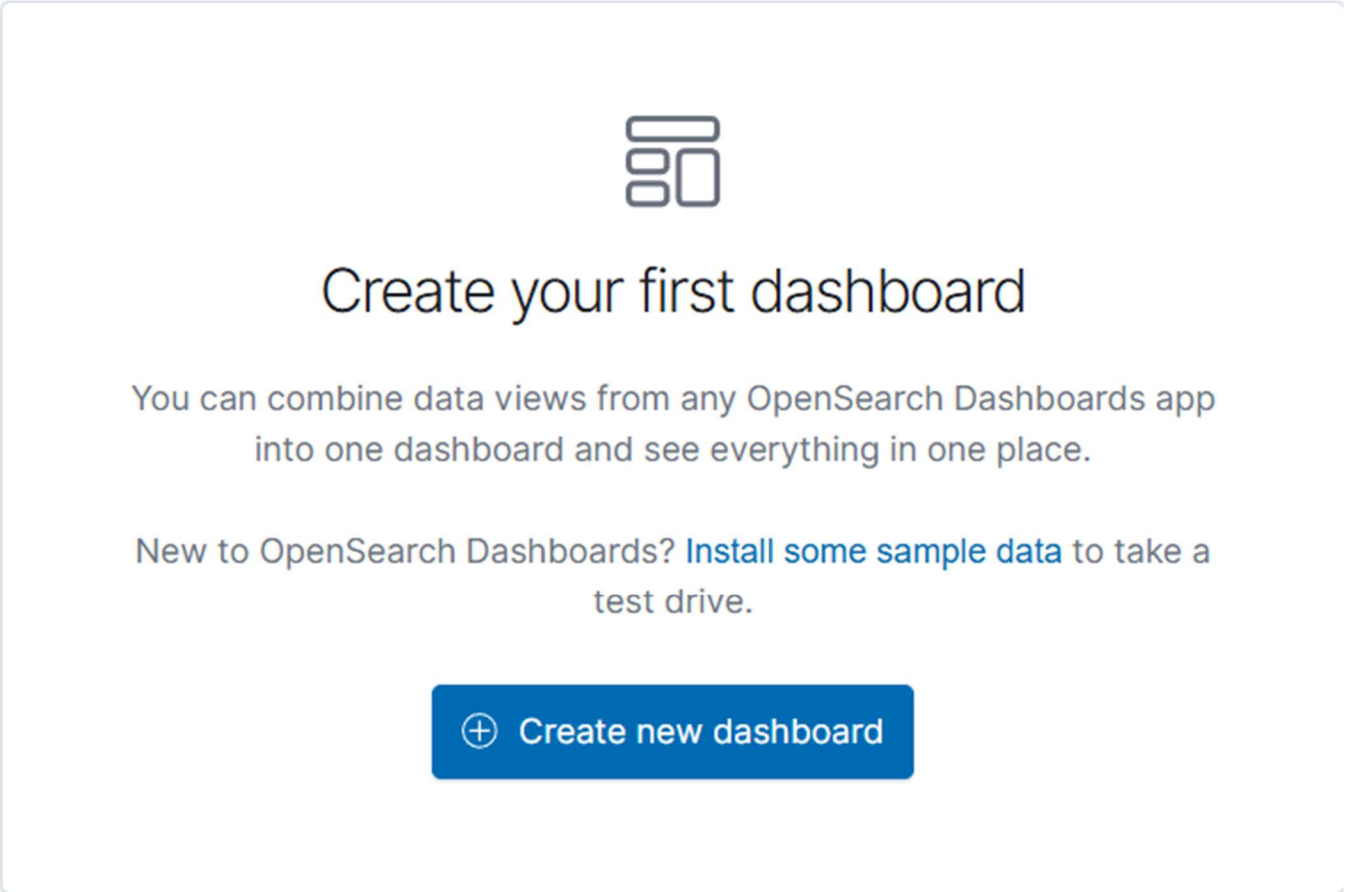
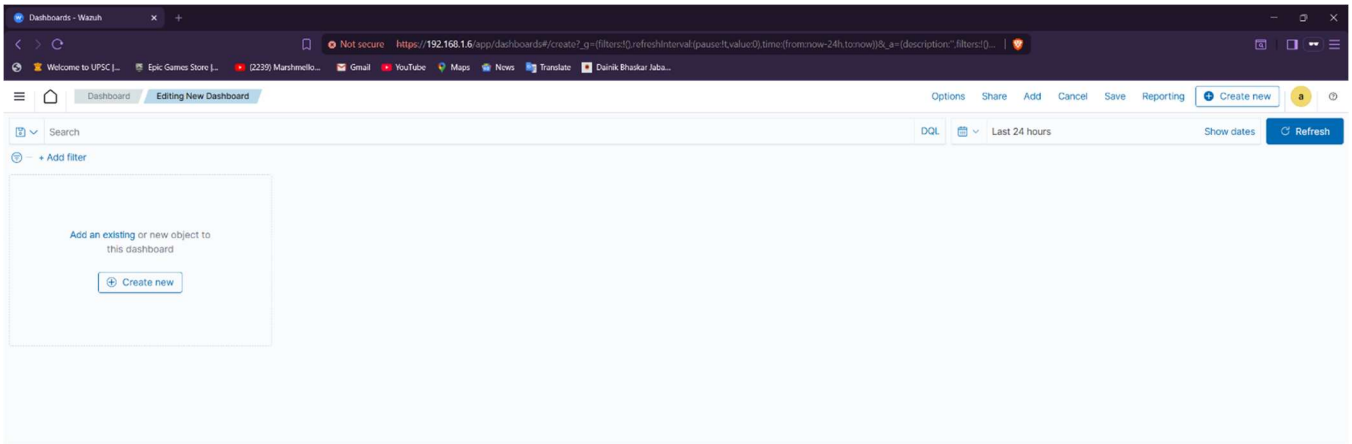
SECURITY INFORMATION MANAGEMENT
Integrity monitoring
AUDITING AND POLICY MONITORING
Policy monitoring
System auditing
Security configuration assessment
THREAT DETECTION AND RESPONSE
MITRE ATT&CK
REGULATORY COMPLIANCE
PCI DSS
NIST 800-53
TSC
GDPR
HIPAA


Create your first dashboard
You can combine data views from any OpenSearch Dashboards app into one dashboard and see everything in one place.
New to OpenSearch Dashboards? Install some sample data to take a test drive.
Create new dashboard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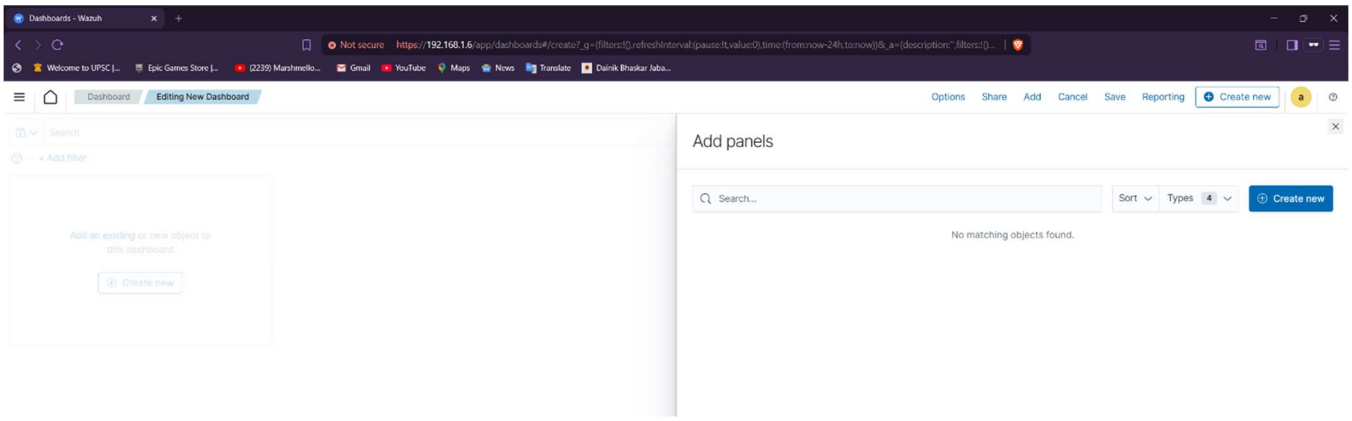
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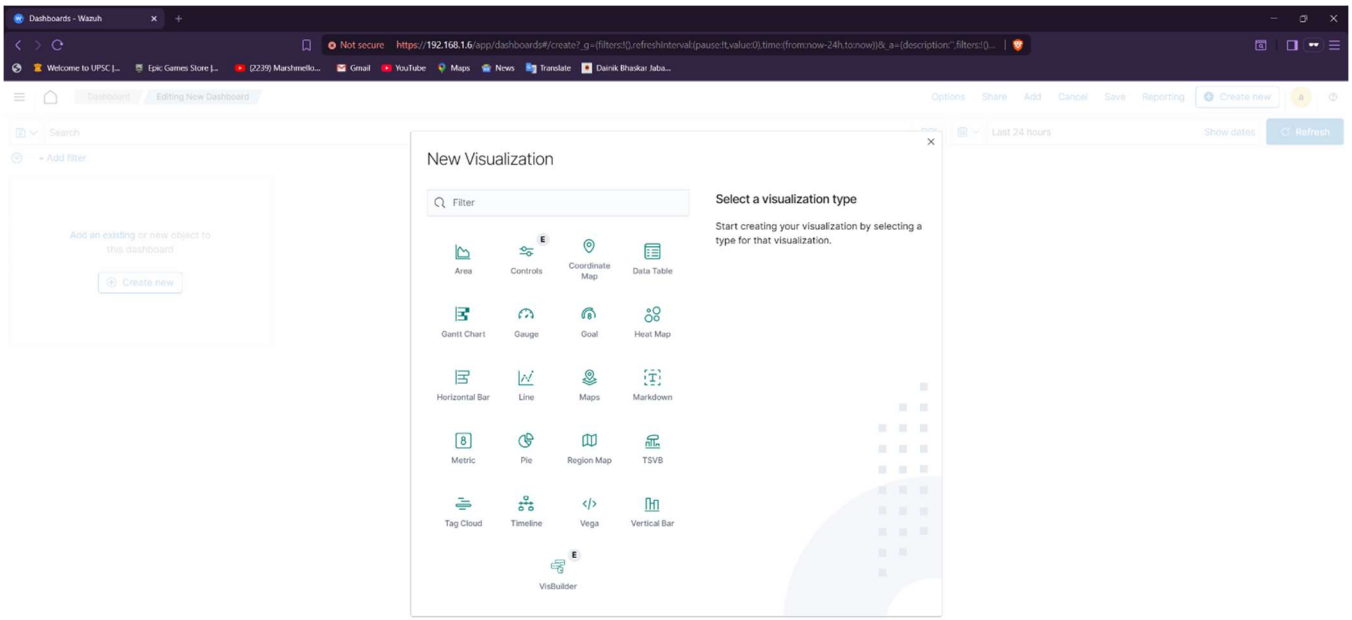
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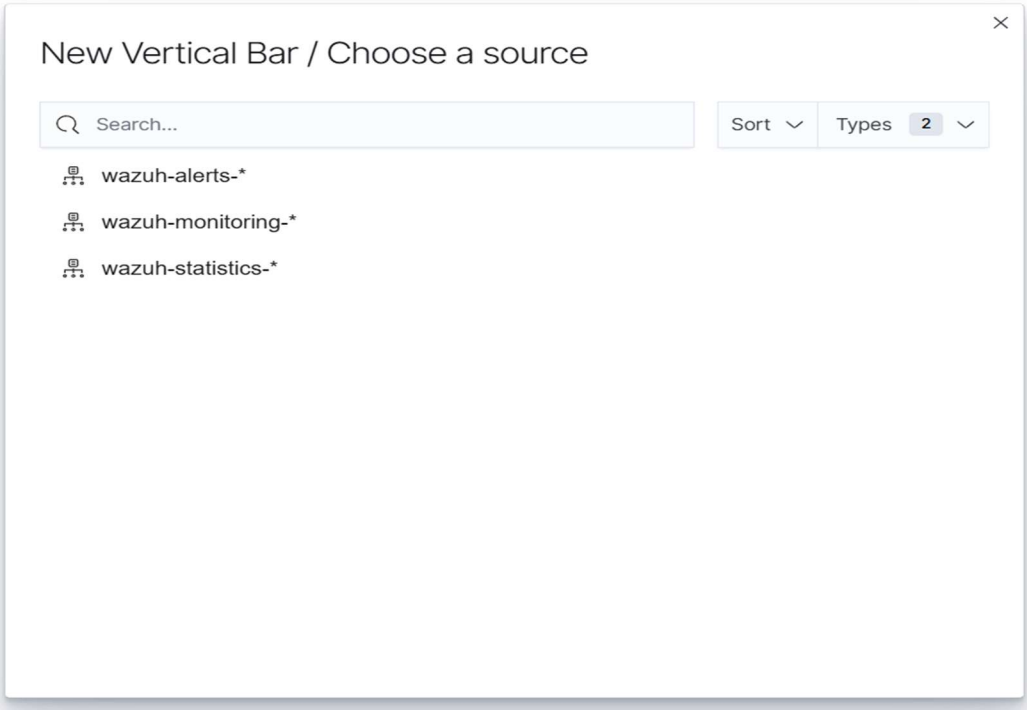
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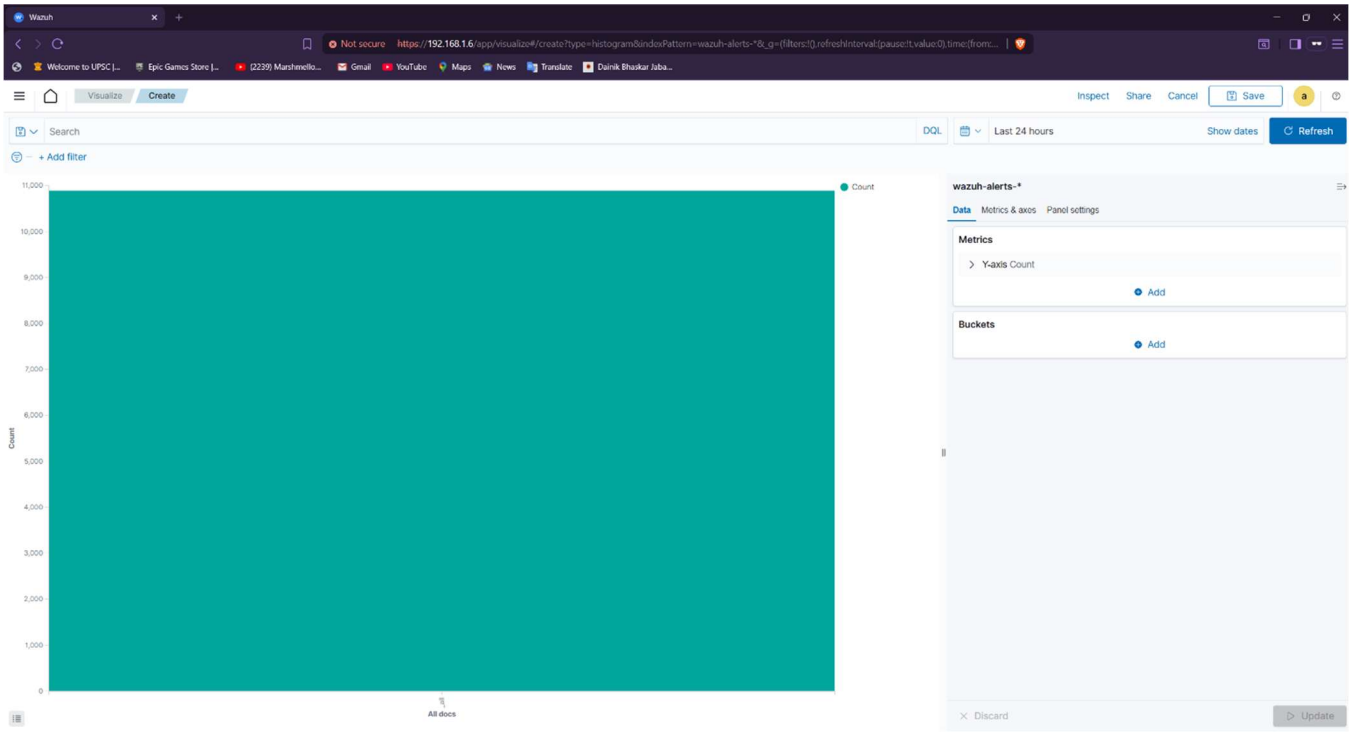
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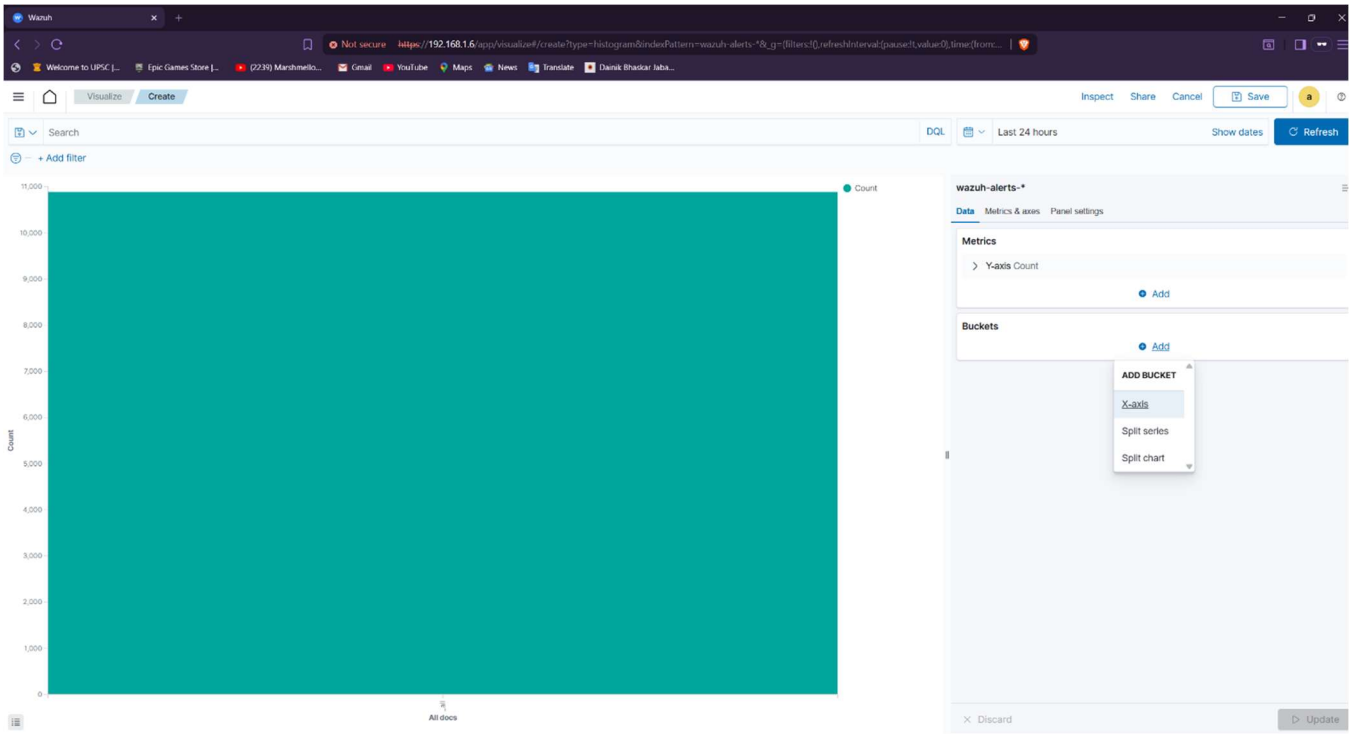
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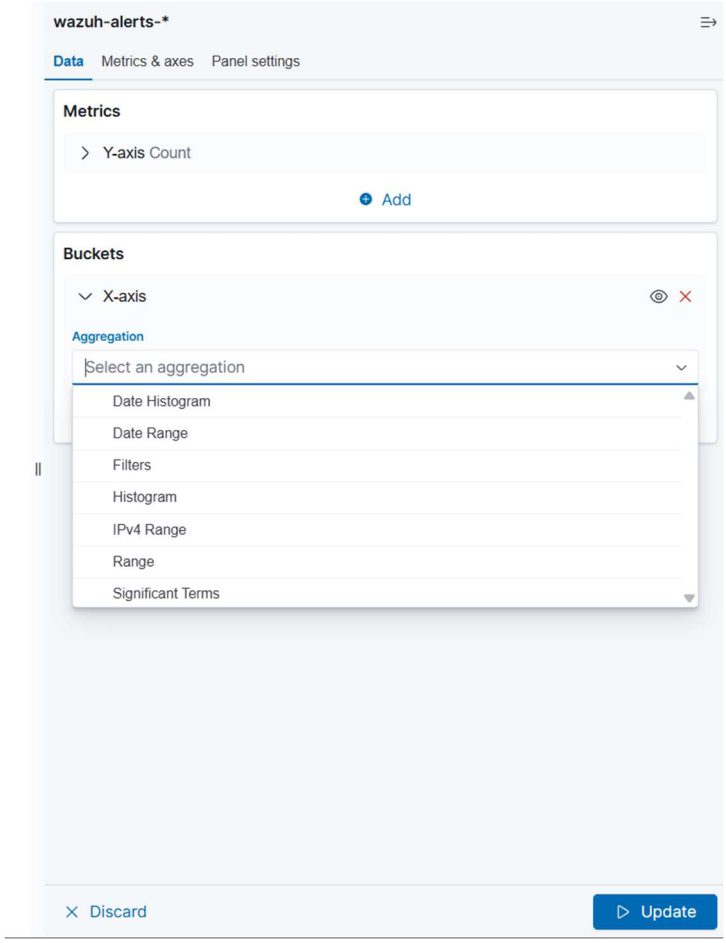
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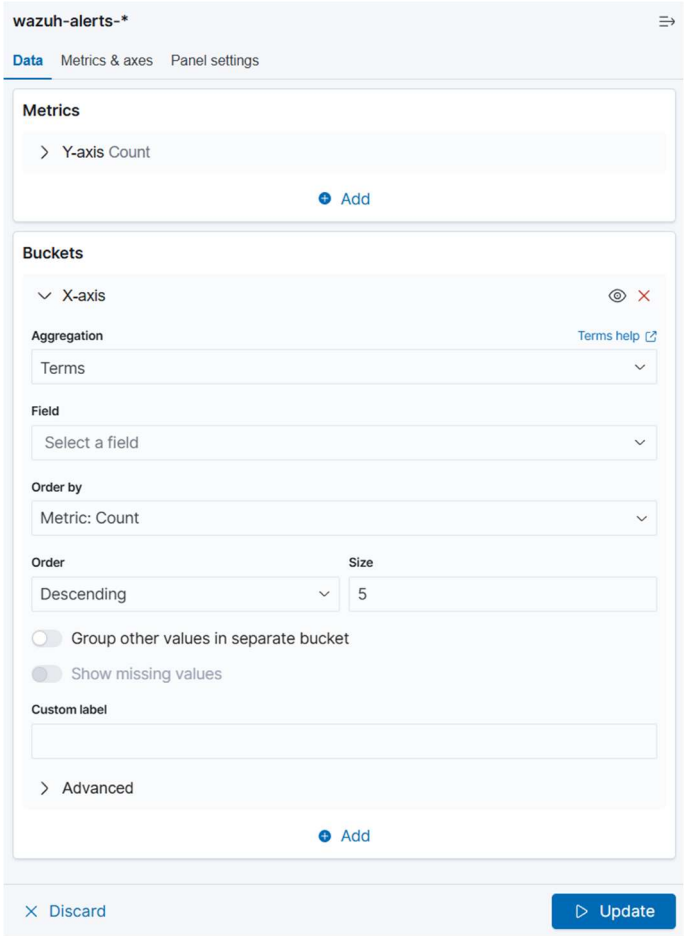
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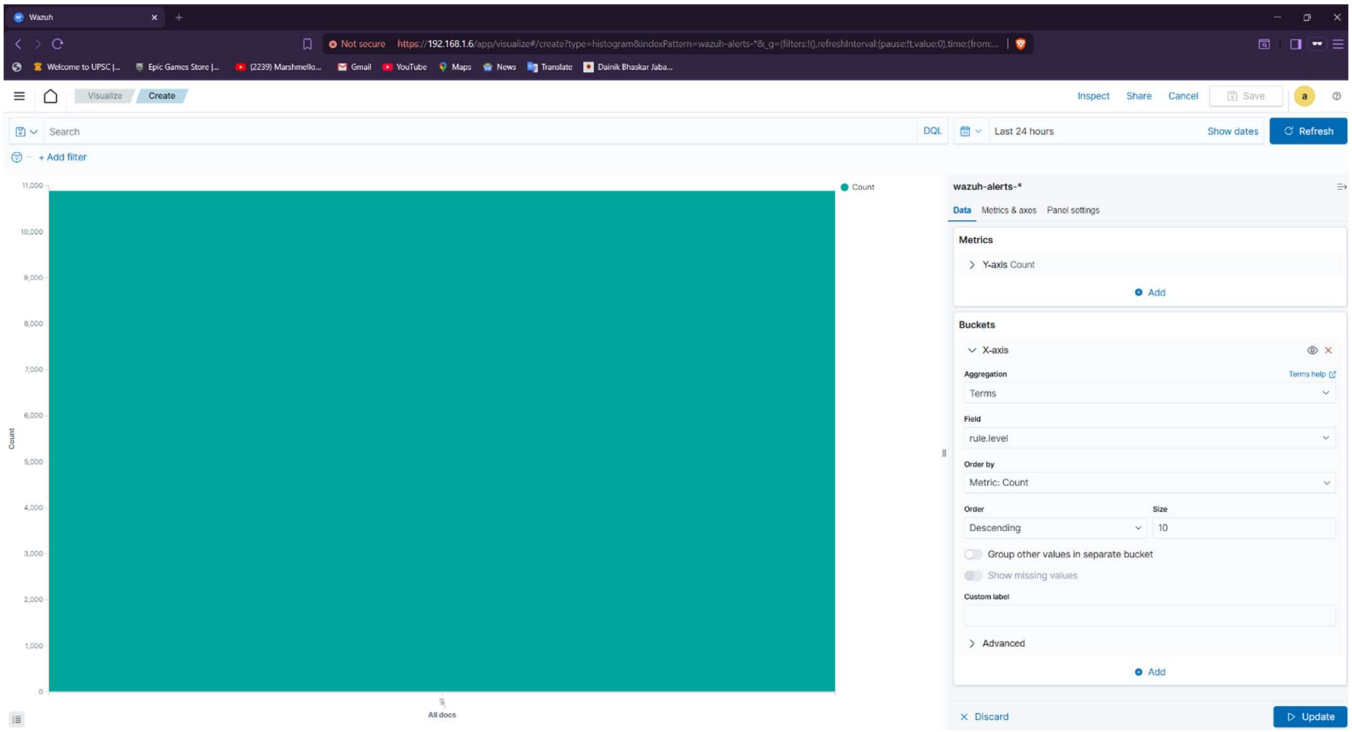
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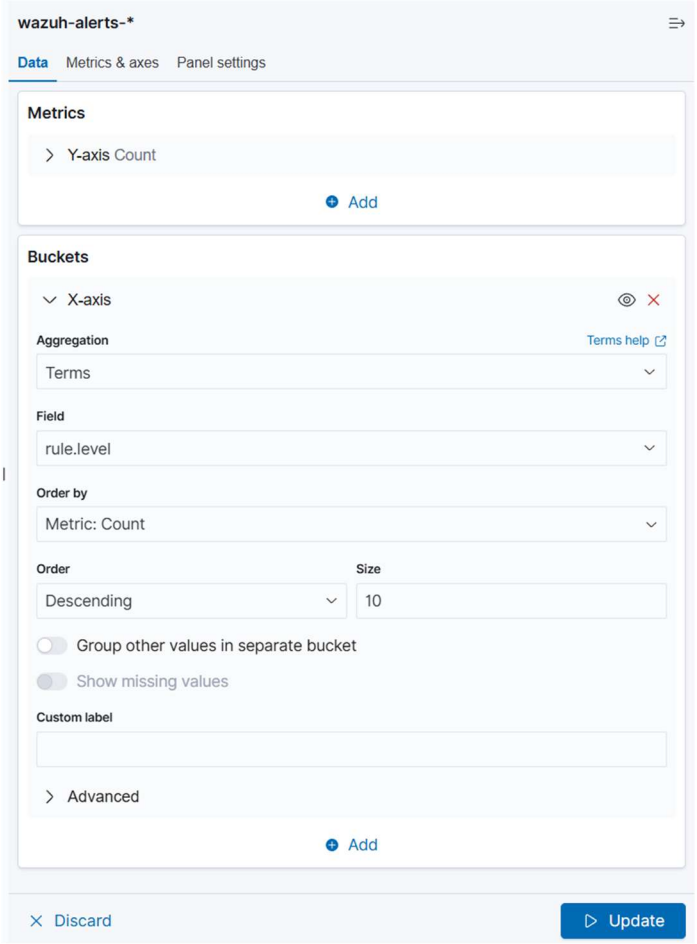
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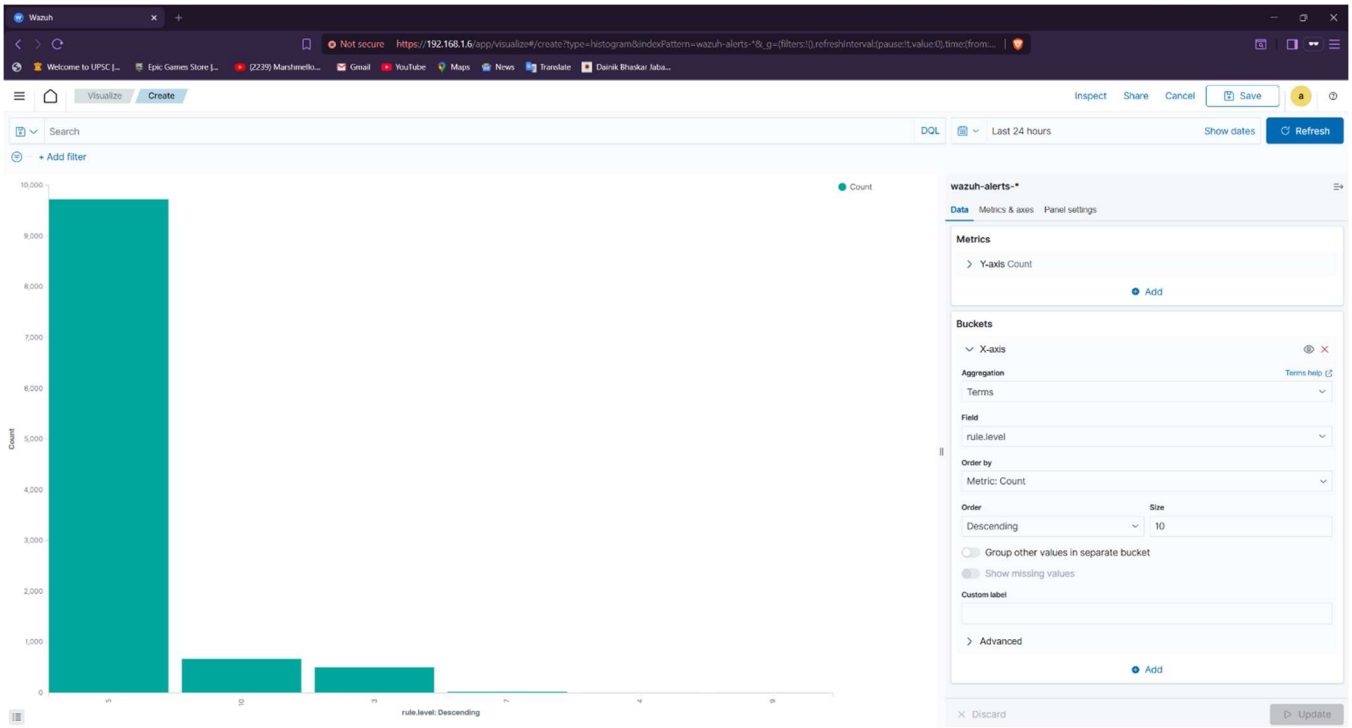
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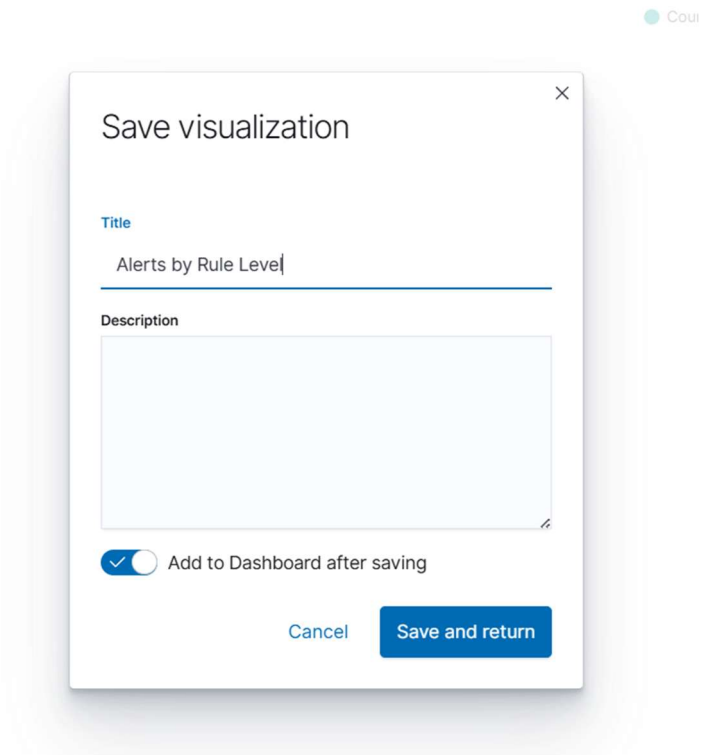
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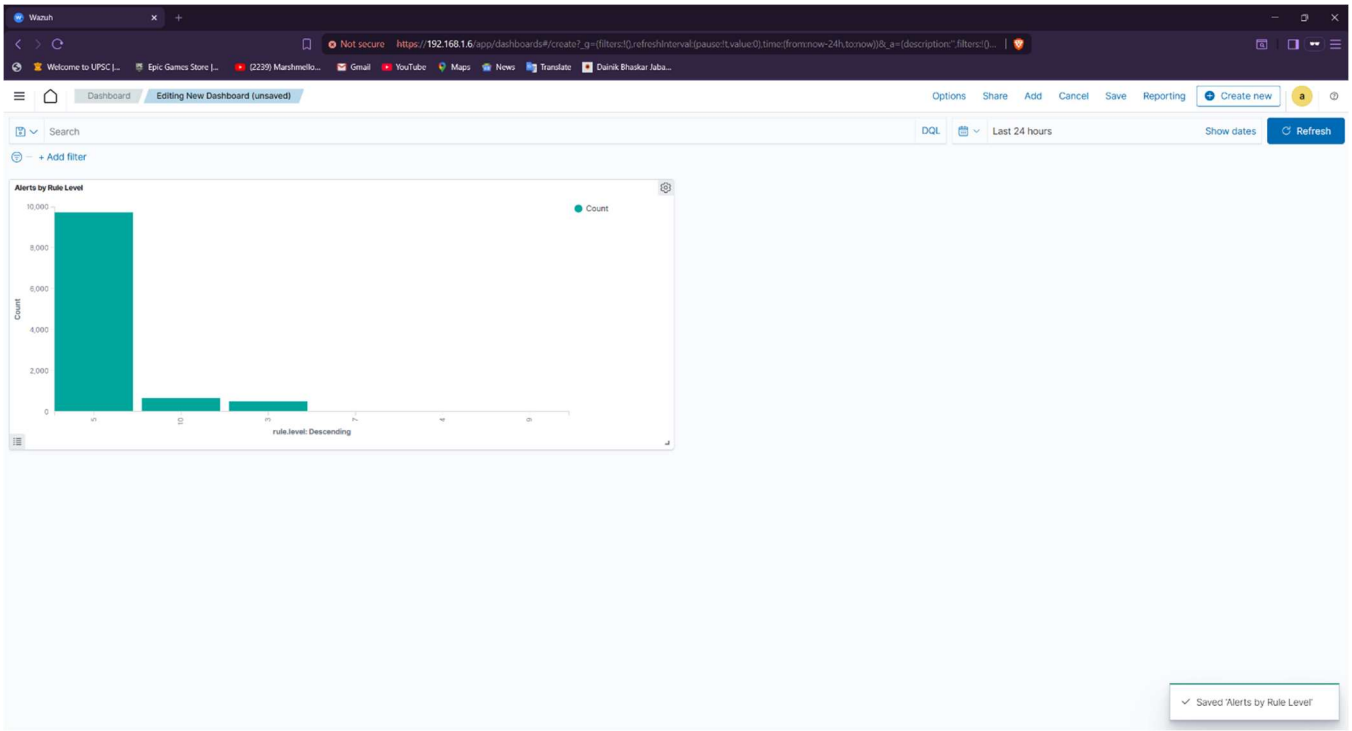
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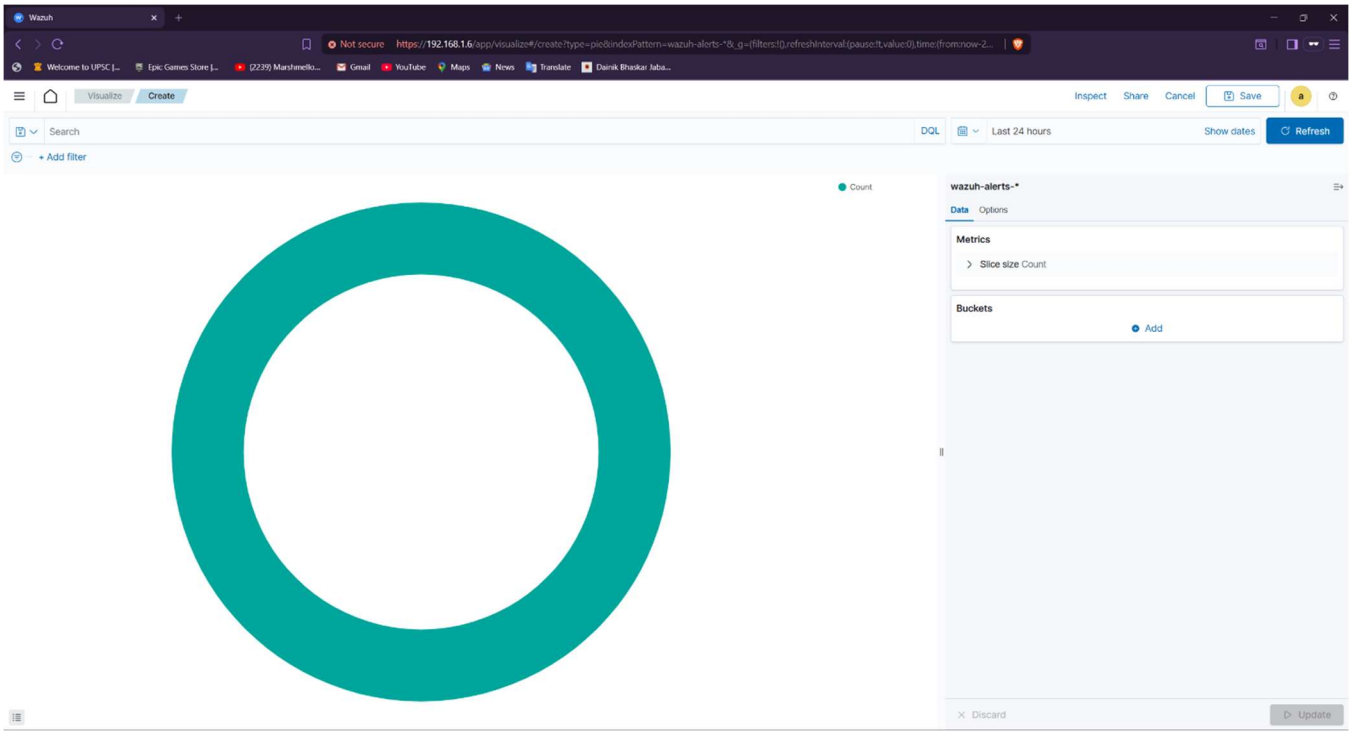
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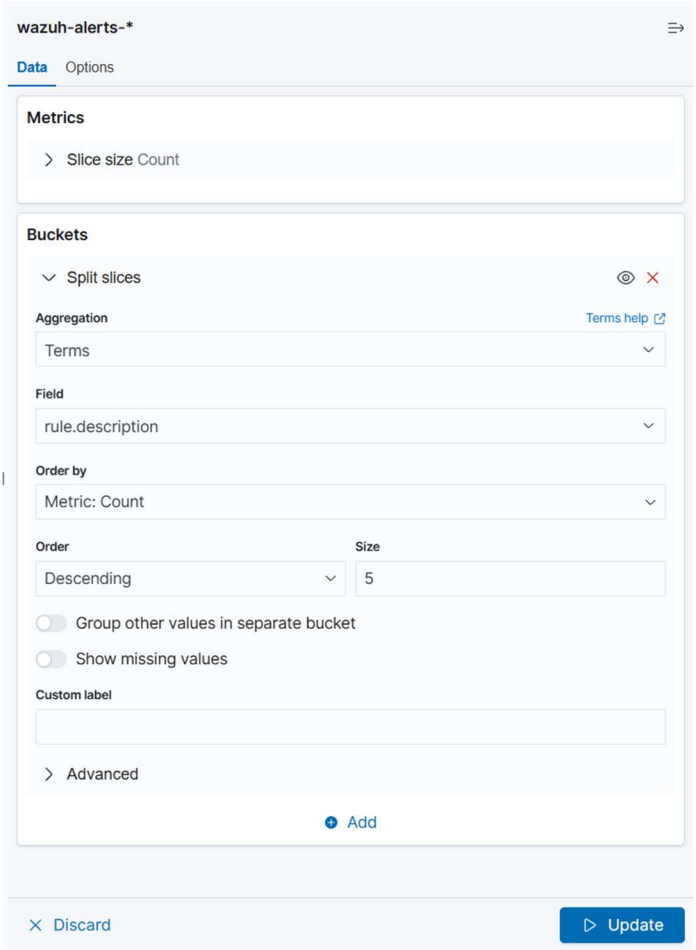
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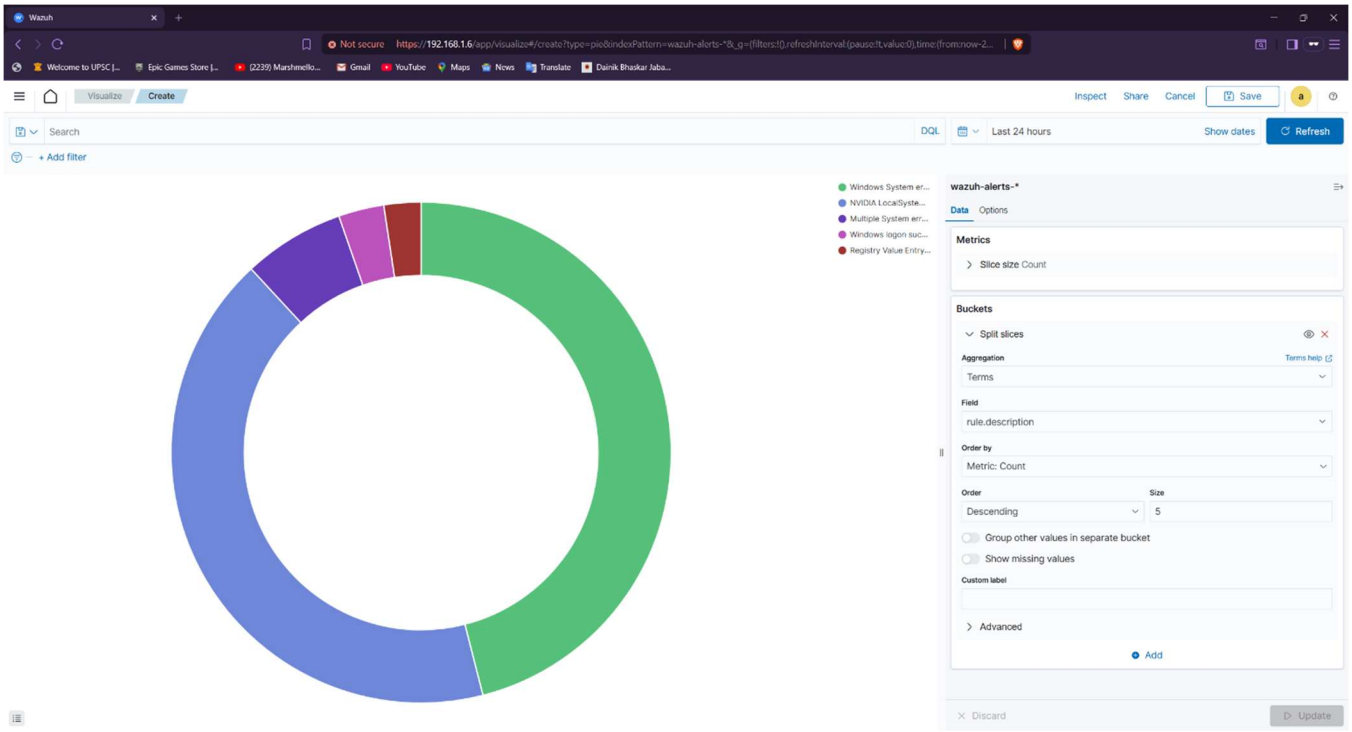
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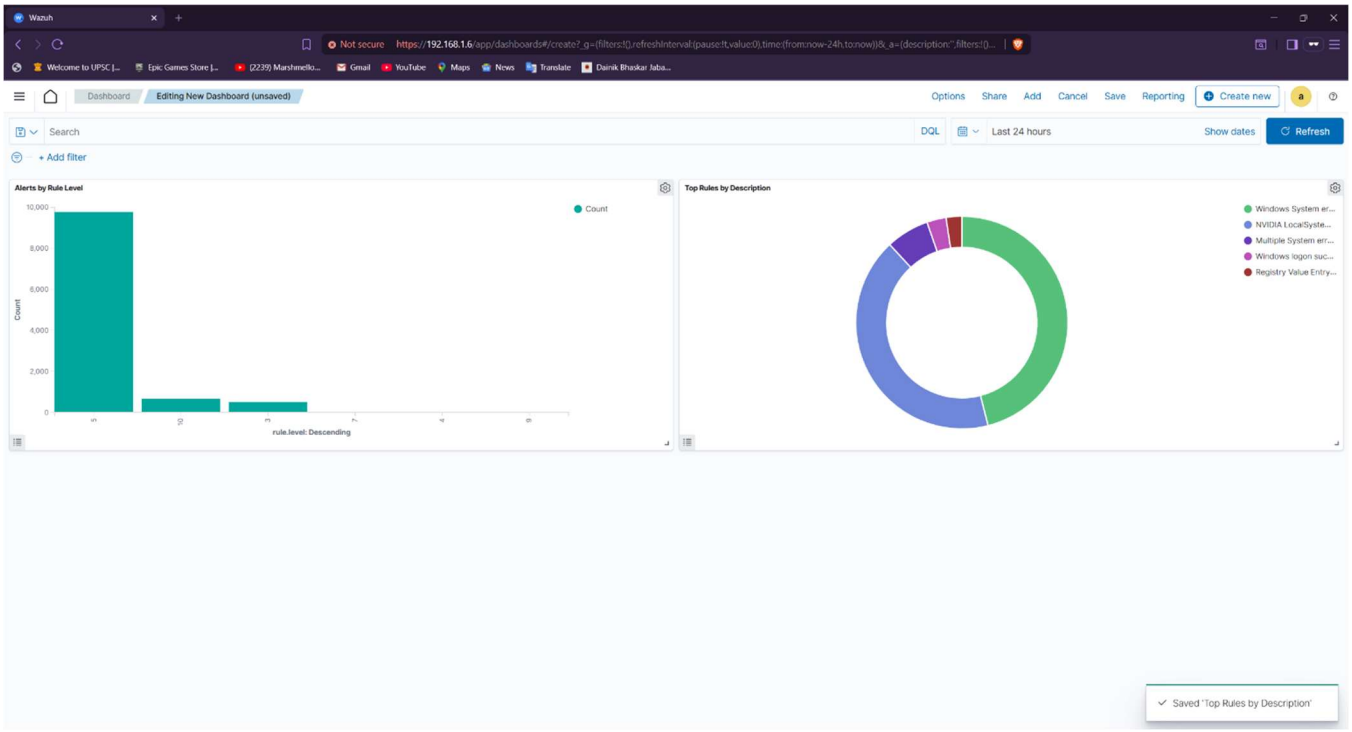
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Dashboard
Editing New Dashboard (unsaved)
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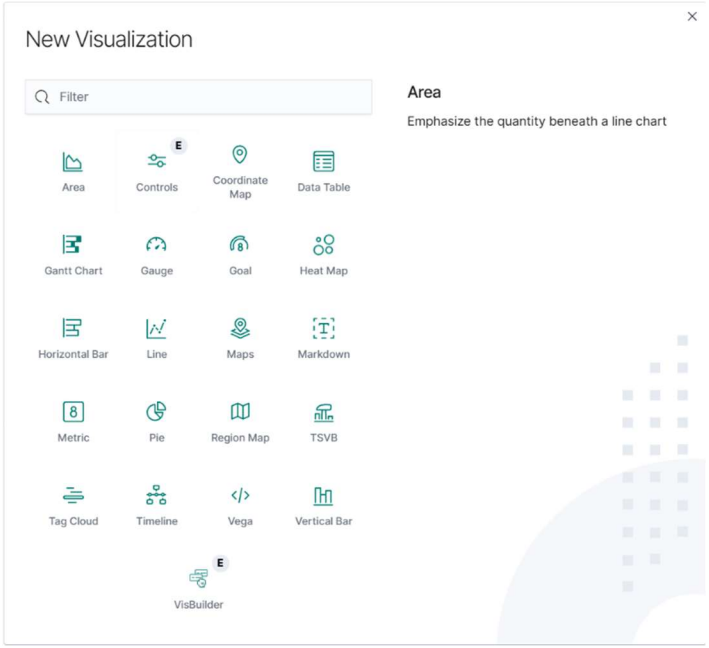
NVIDIA LocalSyste...
Multiple System err...
Windows logon suc...
Registry Value Entry...
Saved 'Top Rules by Description'


New Visualization
Filter
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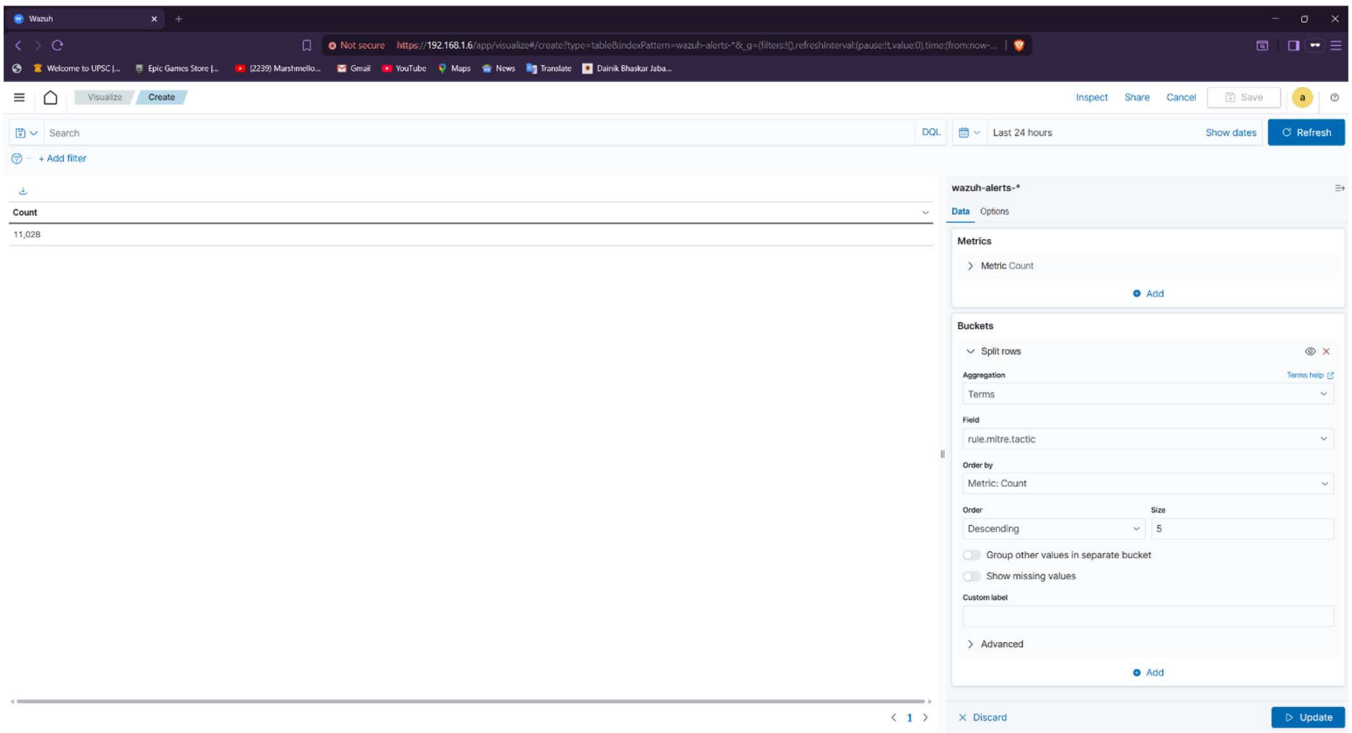
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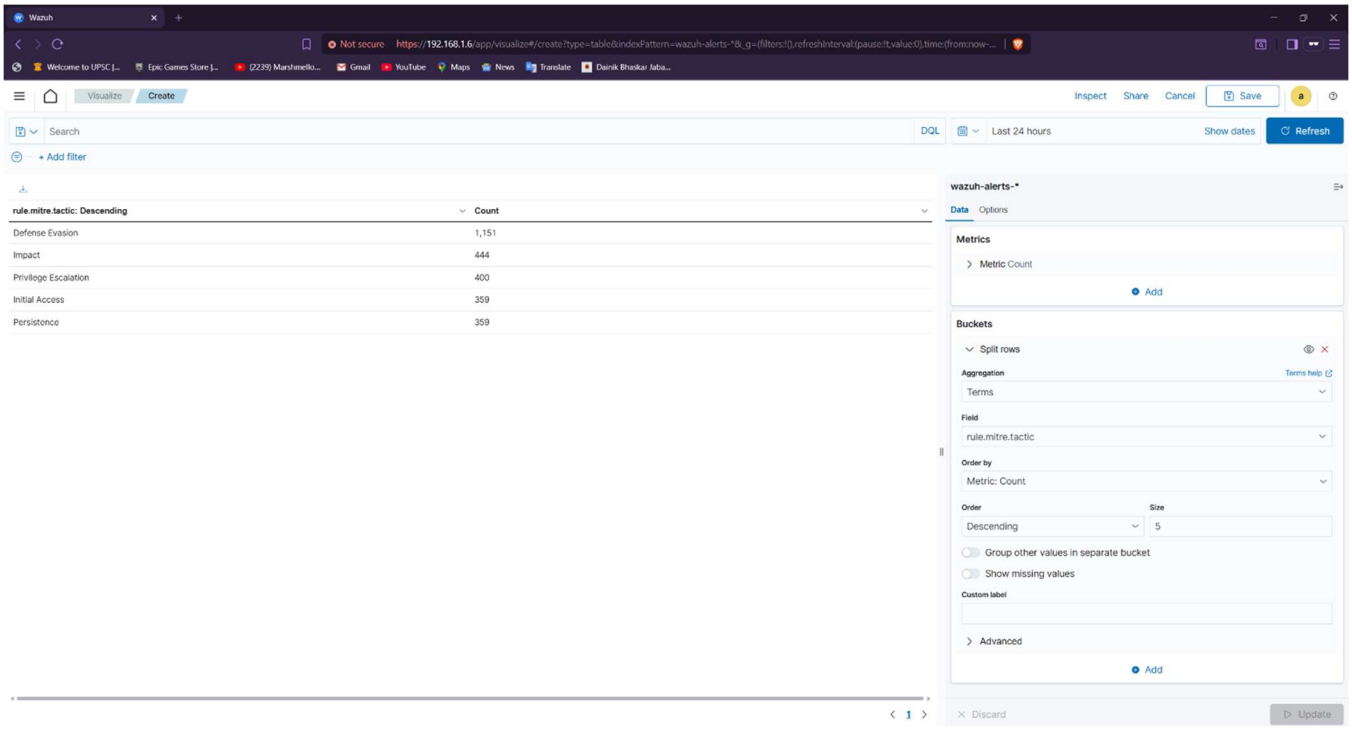
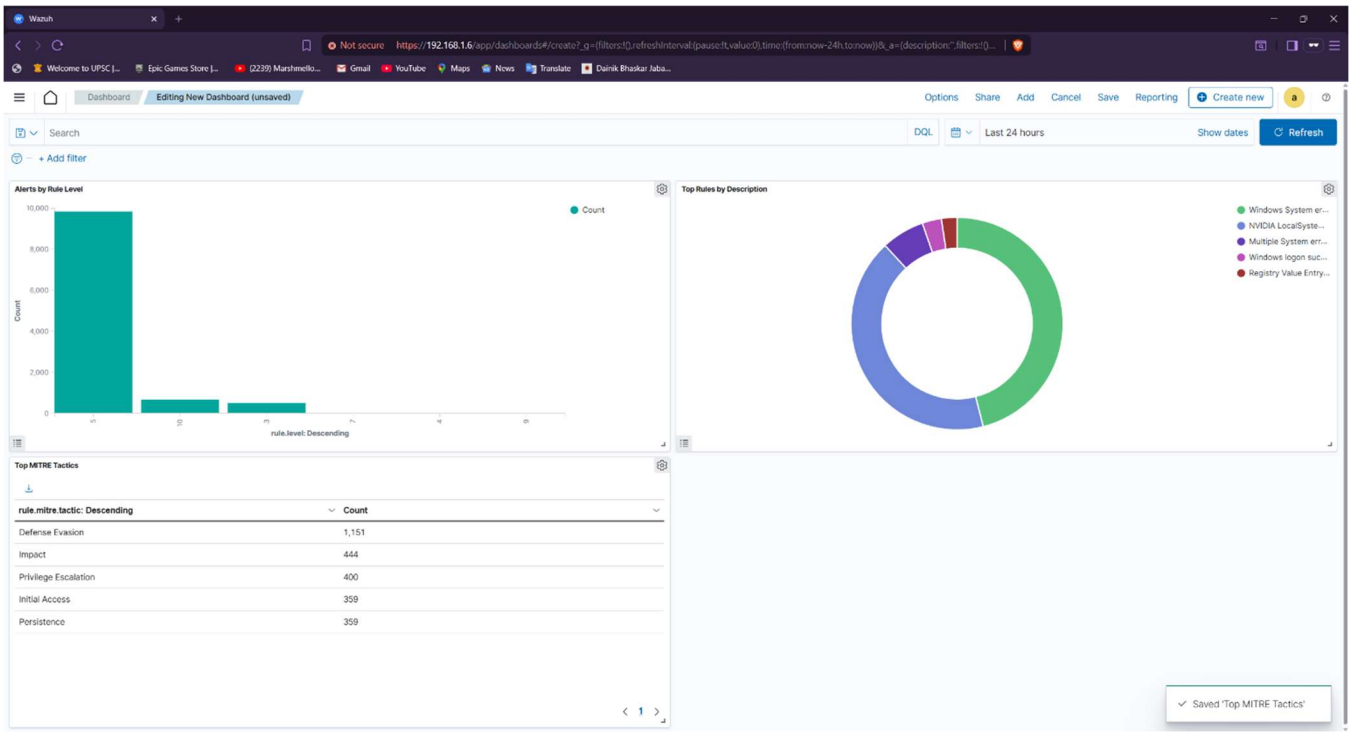
5
Group other values in separate bucket
Show missing values
Custom label
Advanced
Add
Discard
Update


Wazuh
Not secure https://192.168.1.6/app/visualize#/create?type=table&indexPattern=wazuh-alerts-*&_g=(filters:!(),refreshInterval:(pause:!t,value:0),time:(from:now-...
Visualize
Create
Inspect
Share
Cancel
Save
Search
DQL
Last 24 hours
Show dates
Refresh
+ Add filter
rule.mitre.tactic: Descending
Count
Defense Evasion 1,151
Impact 444
Privilege Escalation 400
Initial Access 359
Persistence 359
wazuh-alerts-*
Data
Options
Metrics
Metric Count
Add
Buckets
Split rows
Aggregation
Terms help
Terms
Field
rule.mitre.tactic
Order by
Metric: Count
Order
Size
Descending
5
Group other values in separate bucket
Show missing values
Custom label
Advanced
Add
Discard
Update


Wazuh
Not secure https://192.168.1.6/app/dashboards#/create?_g=(filters:!(),refreshInterval:(pause:!t,value:0),time:(from:now-24h,to:now))&_a=(description:'',filters:!()...
Dashboard
Editing New Dashboard (unsaved)
Options
Share
Add
Cancel
Save
Reporting
Create new
Search
DQL
Last 24 hours
Show dates
Refresh
+ Add filter
Alerts by Rule Level
Count
rule.level: Descending
Top Rules by Description
Windows System er...
NVIDIA LocalSyste...
Multiple System err...
Windows logon suc...
Registry Value Entry...
Top MITRE Tactics
rule.mitre.tactic: Descending
Count
Defense Evasion 1,151
Impact 444
Privilege Escalation 400
Initial Access 359
Persistence 359
Saved 'Top MITRE Tactics'